

LINUX ASSIGNMENT-4

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SECTION:B,ROLL NO:21

QUESTION 1

1. A system has a file /etc/passwd. How would you use grep + tee to extract usernames and save them to a file while also displaying them on screen?

The /etc/passwd file has entries like:

username:x:UID:GID:comment:home:shell

So the **username** is the first field, separated by :.

To extract just the usernames, you can use cut or awk, but since you asked for grep + tee, here's one way:

```
grep -o '^[^:]*' /etc/passwd | tee usernames.txt
```

QUESTION 2

2. A binary isn't found in \$PATH. How would you use commands (which, find, locate) to troubleshoot and fix the issue?

- 1) Check if the binary is available at all

which <binary-name>
- 2) Search the filesystem for the binary

sudo find / -type f -name <binary-name> 2>/dev/null
- Use locate(faster, uses a database-update with sudo updatedb if needed

locate <binary-name>
- 3) Fix the issue depending on the result

export PATH=\$PATH:/usr/local/bin

QUESTION 3

3. Write a command pipeline that finds all .log files modified in the last 24 hours in /var/log and saves results into log_report.txt.

```
find /var/log -type f -name "*.log" -mtime -1 > log_report.txt
```


- `find /var/log` → search under `/var/log`
- `-type f` → only regular files
- `-name "*.log"` → only files ending in `.log`
- `-mtime -1` → modified within the last 24 hours (less than 1 day ago)
- `log_report.txt` → redirect results into `log_report.txt`

QUESTION 4

4. What is the difference between `shutdown -r now` and `reboot`?

Shutdown-r	reboot
shutdown is a more graceful way to bring the system down.	A more direct way to restart the system.
-r means reboot after shutting down.	Essentially tells the kernel to initiate a reboot right away.
It notifies logged-in users (via wall messages), and runs shutdown scripts (<code>/etc/init.d</code> , <code>systemd</code> units, etc.).	On modern Linux (<code>systemd</code> -based), <code>reboot</code> is actually a symlink to <code>systemctl reboot</code> , which behaves similarly to <code>shutdown -r now</code> .

QUESTION 5

5. How can you use the `tee` command to debug a script that generates both standard output and error messages?

Basic usage with both `stdout` and `stderr`:

Bash- `./yourscript.sh 2>&1 | tee debug.log`

If you want to separate `stdout` and `stderr` into different files:

Bash- `./yourscript.sh >>(tee stdout.log) 2>>(tee stderr.log >&2)`

When to use which:

- Combined log → `2>&1 | tee debug.log` (simpler, good for most debugging).
- Separate logs → more advanced redirection (useful if you want to distinguish errors from normal output).

QUESTION 6

6. Explain any three real-world applications of Linux in industries?

- Web Servers & Hosting (IT/Tech Industry)

ASPECT	APPLICATION SOFTWARE	SYSTEM SOFTWARE	UTILITY SOFTWARE
DEFINITION	Programs designed to perform specific tasks for the user.	Software that manages hardware and provides a platform for running applications.	Programs that perform maintenance, optimization, and management tasks for the system.
PURPOSE	To help users accomplish tasks like word processing, browsing, or multimedia.	To control and manage hardware resources and provide services to applications.	To monitor, maintain, and enhance the performance or security of the system.
EXAMPLES IN LINUX	LibreOffice, Firefox, GIMP, VLC media player	Linux kernel, systemd, X Window System, GNOME/KDE desktop environment	top, htop, df, fsck, cron, tar, gzip
USER INTERACTION	Direct interaction with the user.	Mostly runs in the background; indirect interaction.	Can be run by users or automatically; provides system feedback and control.

- Embedded Systems & IoT (Automotive, Consumer Electronics)
- Supercomputing & Scientific Research
- Telecommunications

QUESTION 7

7. Differentiate application, system and utility software in the context of linux environment?

QUESTION 8

8. What are the key differences between open-source and proprietary operating systems?

ASPECT	OPEN-SOURCE OS	PROPRIETARY OS
DEFINITION	The source code is publicly	The source code is closed;

	available; users can view, modify, and distribute it.	only the vendor can modify or distribute it.
COST	Usually free or low-cost	Requires purchasing a license, often expensive

QUESTION 9

9. Write the command to display the system's kernel version.

1) Using uname:

uname -r

QUESTION 10

10. What is the difference between head and tail commands in text processing?

ASPECT	HEAD	TAIL
PURPOSE	Displays the beginning of a file or input.	Displays the end of a file or input.
DEFAULT BEHAVIOUR	Shows the first 10 lines of a file.	Shows the last 10 lines of a file.